

Notes: Borusyak, Hull, and Jaravel (RES, 2022)

Quasi-Experimental Shift-Share Research Designs

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1 One-sentence summary

Borusyak, Hull, and Jaravel show that shift-share IV regressions can be understood as quasi-experiments in the underlying shocks, derive an equivalent shock-level regression, and use this equivalence to guide identification, inference, balance tests, and robustness checks.

2 Big picture

2.1 Question

Many empirical papers use shift-share, or Bartik, instruments of the form

$$z_\ell = \sum_n s_{\ell n} g_n,$$

where unit ℓ is exposed to shocks g_n through shares $s_{\ell n}$. The paper asks: what assumptions justify these instruments, how should researchers conduct inference, and how should researchers diagnose whether the identifying variation is credible?

2.2 Main answer

The paper develops a quasi-experimental view of shift-share IV. Identification can come from quasi-random assignment of the shocks g_n , even when exposure shares $s_{\ell n}$ are endogenous. The key is to move from the observation level to an equivalent shock-level regression.

2.3 Why this matters

In many applications, shares are historical exposure variables such as industry employment shares, immigrant-origin shares, or export-market shares. These shares are rarely random. The paper provides a way to justify shift-share designs without pretending shares are exogenous, by putting the quasi-experiment at the shock level.

2.4 Main empirical illustration

The paper re-analyzes the Autor, Dorn, and Hanson China-shock setting. It studies whether Chinese import-growth shocks are plausibly quasi-random at the industry level after conditioning on appropriate shock-level controls, and it shows how estimates change under shock-level diagnostics and robustness checks.

3 Incremental contribution of the paper

3.1 Equivalence result

The central contribution is a formal equivalence: the conventional shift-share IV estimator is equal to a shock-level IV estimator using exposure-weighted aggregates. This transforms a regional IV design into a shock-level quasi-experiment.

3.2 New interpretation of Bartik-style instruments

The paper clarifies that shift-share validity can come from either share exogeneity, shock exogeneity, or a mixture. Its focus is the shock-exogeneity case: shares may be endogenous, but shocks are quasi-random conditional on observables.

3.3 Practical diagnostics

The framework gives concrete diagnostics: report shock-level Herfindahl concentration, test shock balance against predetermined shock-level or exposure-weighted characteristics, control for expected shock means, and use shock-level standard errors.

3.4 New inference guidance

The paper shows that conventional observation-level standard errors may be inappropriate when the quasi-experiment is at the shock level. It motivates exposure-robust and shock-level inference based on the equivalent regression.

3.5 A bridge between two literatures

The paper connects applied shift-share designs with econometric work on many-instrument asymptotics, clustered shocks, random-assignment inference, and quasi-experimental balance testing.

4 Data and measurement

4.1 Generic shift-share objects

The basic shift-share instrument is

$$z_\ell = \sum_{n=1}^N s_{\ell n} g_n.$$

Here ℓ indexes units such as regions, firms, or individuals; n indexes shocks such as industries, countries, demographic groups, or markets; $s_{\ell n}$ is unit ℓ 's exposure to shock n ; and g_n is the shock.

Interpretation: The exposure shares describe who is exposed to what. The shocks are the source of identifying variation in the quasi-experimental interpretation.

4.2 SSIV regression

The canonical shift-share IV equation is

$$y_\ell = \beta x_\ell + w'_\ell \gamma + \varepsilon_\ell,$$

with x_ℓ instrumented by z_ℓ . The IV moment is

$$E[z_\ell \varepsilon_\ell] = 0.$$

Interpretation: The paper asks when this observation-level moment can be justified by assumptions on the underlying shocks.

4.3 Shock-level aggregates

Define exposure-weighted shock-level aggregates, for example

$$\bar{y}_n = \frac{\sum_\ell e_\ell s_{\ell n} y_\ell}{s_n}, \quad s_n = \sum_\ell e_\ell s_{\ell n},$$

where e_ℓ are observation weights. Similar aggregates are defined for x_ℓ , controls, and residuals.

Interpretation: These aggregates are the key bridge. They map unit-level variables into shock-level variables weighted by the exposure shares that make each shock relevant.

4.4 Effective shock sample size

The paper emphasizes the exposure-weight Herfindahl index:

$$HHI = \sum_n s_n^2, \quad N_{eff} = 1/HHI.$$

Interpretation: The effective sample size can be far smaller than the raw number of shocks if exposure is concentrated in a few shocks. Reporting $1/HHI$ is a simple diagnostic for whether the shock-level quasi-experiment has enough independent variation.

5 Core econometric framework

5.1 Proposition 1: equivalent shock-level regression

The paper’s equivalence result states that the conventional SSIV estimator equals the second-stage coefficient from a shock-level IV regression, with shock weights s_n and appropriately aggregated outcomes, treatments, and controls.

$$\hat{\beta}_{SSIV} = \hat{\beta}_{shock-level}.$$

Interpretation: This proposition is the paper’s organizing device. Once the estimator is rewritten at the shock level, identification and inference can be discussed using shock-level quasi-random assignment.

5.2 Shock orthogonality

The observation-level orthogonality condition can be represented as a shock-level condition:

$$E \left[\sum_n s_n g_n \bar{\varepsilon}_n \right] = 0.$$

If shocks are quasi-randomly assigned relative to the shock-level unobservable $\bar{\varepsilon}_n$, the instrument is valid.

Interpretation: Instead of asking whether regions with different exposure shares are comparable, the paper asks whether industries or shocks with different g_n values are comparable after conditioning.

5.3 Baseline assumptions

The baseline quasi-experimental framework uses two key assumptions:

- **Quasi-random shock assignment:** shocks have common conditional expectation after controlling for shock-level observables.
- **Many uncorrelated shocks:** no small set of shocks dominates the identifying variation, and shocks are mutually uncorrelated or weakly dependent.

Interpretation: These assumptions are shock-level analogues of familiar experimental requirements: balance and enough independent treatment variation.

5.4 Conditional shock assignment

If shocks have different expected values across groups or periods, researchers can control for shock-level variables q_n . In the observation-level regression this means controlling for the corresponding shift-share variables:

$$\sum_n s_{\ell n} q_n.$$

Interpretation: This is one of the paper’s most practical insights: if shock means differ by period, sector, or cluster, the observation-level regression should include exposure-weighted controls for those shock categories.

5.5 Incomplete shares

If exposure shares do not sum to one, the omitted share can be treated as exposure to a missing shock. If that missing shock is mechanically zero, it may create identification problems unless researchers control for total exposure or restrict attention to relevant shocks.

Interpretation: In the ADH application, including service industries with a zero shock creates severe concentration and violates the quasi-random shock-assignment interpretation. Excluding them is more appropriate.

6 Inference and testing

6.1 Exposure-robust standard errors

Because the equivalent regression is at the shock level, standard errors should generally reflect shock-level variation. The paper provides conditions under which conventional shock-level standard errors are valid or conservative.

Interpretation: If the quasi-experiment is that industries receive quasi-random shocks, treating thousands of regions as independent observations can overstate precision.

6.2 Falsification tests

Researchers can test whether shocks are correlated with predetermined shock-level or exposure-weighted covariates. A typical test regresses a predetermined characteristic on the shocks using the equivalent shock-level regression.

Interpretation: These balance tests do not prove exogeneity, but they discipline the research design in the same way covariate balance checks discipline randomized or quasi-random designs.

6.3 First-stage relevance tests

The equivalent shock-level regression also allows researchers to test first-stage relevance at the shock level. This avoids relying only on observation-level first-stage statistics that may be inflated by exposure structure.

Interpretation: A strong observation-level first stage may still come from too few effective shocks. Shock-level first-stage diagnostics are therefore central.

7 Application: Autor, Dorn, and Hanson China shock

7.1 Empirical setting

The empirical application revisits the ADH estimate of the effect of Chinese import competition on U.S. manufacturing employment across commuting zones. The instrument is a shift-share measure of import-growth shocks by industry, weighted by local industry employment shares.

7.2 Identifying variation

The quasi-experimental view treats Chinese import growth in different industries as shocks. The key question is whether these industry shocks are plausibly quasi-random after conditioning on period and industry-level observables.

7.3 Important practical changes

The paper shows that the nonmanufacturing service sector should be excluded from the shock set because it has a zero shock and enormous exposure weight. Excluding it dramatically increases the effective shock sample size.

8 Identification and empirical design

8.1 What the paper identifies

The paper identifies conditions under which an SSIV coefficient has a causal interpretation when the identifying variation comes from shocks rather than shares. It identifies the coefficient using cross-shock variation in g_n , weighted by average exposure.

8.2 What can be endogenous

Exposure shares $s_{\ell n}$ can be correlated with regional unobservables. This is allowed, provided the shocks are quasi-random conditional on the shock-level and exposure-weighted controls used in the design.

8.3 Balance logic

Balance tests should be conducted at the shock level. In the ADH application, the paper tests whether industry-level shocks are related to predetermined industry characteristics and whether regional exposure to shocks is related to regional characteristics.

8.4 Sensitivity logic

Researchers should show how estimates change when they remove problematic shock variation, residualize shocks on period effects, control for shock-level categories, and add exposure-weighted controls.

8.5 Main limitations

The framework does not automatically validate every shift-share design. It provides a way to make the assumptions explicit. Researchers still need substantive arguments that shocks are quasi-random and sufficiently independent after conditioning.

9 Tables and figures

9.1 Table 1: Shock summary statistics in the Autor et al. (2013) setting; Table 2: Shock intra-class correlations in the Autor et al. (2013) setting

Read: Table 1 reports the distribution of China import-growth shocks and the effective sample size of the exposure-weighted shocks. Table 2 reports intra-class correlations by sector, SIC2, SIC3, and industry.

Economic message: Table 1 shows why service industries should be excluded: including a zero service shock produces a tiny effective shock sample size. Table 2 shows that shocks have nontrivial within-industry and within-group correlation, motivating clustered or shock-level inference.

Interpretation: The key number in Table 1 is the inverse Herfindahl index. When services are included, the effective sample size across industry-periods is only 3.5; after restricting to manufacturing, it becomes 191.6. This changes the credibility of the quasi-experimental interpretation.

TABLE 1
Shock summary statistics in the *Autor et al. (2013)* setting

	(1)	(2)	(3)
Mean	1.79	7.37	0
Standard deviation	10.79	20.92	20.44
Interquartile range	0	6.61	6.11
Specification			
Excluding service industries		✓	✓
Residualizing on period FE			✓
Effective sample size ($1/HHI$ of s_{it} weights)			
Across industries and periods	3.5	191.6	191.6
Across SIC3 groups	1.7	58.4	58.4
Largest s_{it} weight			
Across industries and periods	0.398	0.035	0.035
Across SIC3 groups	0.757	0.066	0.066
Observation counts			
No. of industry-period shocks	796	794	794
No. of industries	398	397	397
No. of SIC3 groups	137	136	136

Notes: This table summarizes the distribution of China import shocks g_{it} across industries n and periods t in the *Autor et al. (2013)* application. Shocks are measured as the total flow of imports from China in eight developed economies outside of the U.S. All statistics are weighted by the average industry exposure shares s_{it} ; shares are measured from lagged manufacturing employment, as described in Section 6.2.1. Column 1 includes the non-manufacturing industry aggregate in each period with a shock of zero, while columns 2 and 3 restrict the sample to manufacturing industries. Column 3 residualizes manufacturing shocks on period indicators. We report the effective sample size (the inverse renormalized Herfindahl index of the s_{it} weights, as described in Section 6.2.2) with and without the non-manufacturing industry, at the industry-by-period level and at the level of SIC3 groups (aggregated across periods), along with the largest s_{it} .

TABLE 2
Shock intra-class correlations in the *Autor et al. (2013)* setting

	Estimate (1)	SE (2)
Shock ICCs		
10 sectors	0.016	(0.022)
SIC2	0.047	(0.052)
SIC3	0.073	(0.057)
Industry	0.169	(0.047)
Period means		
1990s	4.65	(1.38)
2000s	16.87	(3.34)
No. of industry-periods		794

Notes: This table reports intra-class correlation coefficients for the *Autor et al. (2013)* manufacturing shocks, estimated from the hierarchical model described in Section 6.2.2. Estimates come from a maximum likelihood procedure with an exchangeable covariance structure for each industry and sector random effect and with period fixed effects. Robust standard errors are reported in parentheses.

(a) **Table 1: Shock summary statistics.** Read this as a diagnostic for how much shock-level variation actually identifies the SSIV estimate.

inference that takes into account the inherent dependencies of the data. Second, we regress

(b) **Table 2: Shock intra-class correlations.** Read this as evidence on whether industry shocks can be treated as independent or require clustering.

Tables 1 and 2. These tables are the paper's shock-level diagnostics: concentration and dependence.

9.2 Table 3: Shock balance tests in the Autor et al. (2013) setting; Table 4: Shift-share IV estimates of the effect of Chinese imports on manufacturing employment

Read: Table 3 reports balance tests for industry-level and regional-level predetermined variables. Table 4 reports SSIV estimates under alternative controls and shock treatments.

Economic message: Table 3 asks whether shocks look quasi-random after conditioning. Table 4 shows that the ADH estimate is robust in sign and economically important, but its magnitude and precision vary with the quasi-experimental specification.

Interpretation: In Table 3, some balance failures suggest that researchers should be cautious and condition on relevant variables. In Table 4, column 1 reproduces the original style of ADH specification, while later columns isolate within-manufacturing variation, add period-specific controls, add sector controls, and compare

11.3 Practical implications

The paper provides a checklist for future work: define the shocks; calculate effective shock sample size; test balance at the shock level; condition on shock-level observables when needed; use shock-level inference; and conduct sensitivity checks that remove problematic or nonexperimental shock variation.

11.4 Limitations

The framework does not make a shift-share design credible automatically. If shocks are themselves endogenous, if effective sample size is small, or if shocks are strongly correlated in unmodeled ways, the quasi-experimental interpretation may fail.

11.5 Takeaway

The main lesson is that a shift-share instrument is only as credible as the quasi-experiment embedded in its shocks. The paper gives researchers the algebra and diagnostics needed to evaluate that claim.